

# David Le

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## EDUCATION

### University of Massachusetts Boston

B.S., Computer Science | GPA: 3.78 / 4.00

Boston, MA

Expected May 2027

- **Relevant Coursework:** Applied Machine Learning, Industry Applied AI Lab, Computer Security, Database Management, Introduction to Software Engineering

## TECHNICAL SKILLS

- **Machine Learning:** scikit-learn, XGBoost, PyTorch, logistic regression, gradient boosting, random forests, feature engineering, model calibration, hyperparameter tuning
- **LLM and Retrieval:** LangGraph, RAG, ChromaDB, FAISS, embeddings, hybrid search, BM25, reranking, prompt engineering, LLM-as-judge evaluation, multimodal vision
- **Evaluation:** benchmark design, ROC-AUC, log loss, walk-forward validation, bootstrap confidence intervals, ablation studies, data leakage detection
- **Languages:** Python, SQL, Java, JavaScript, C
- **Backend and Infra:** FastAPI, Flask, Docker, PostgreSQL, Kafka, Redis, GitHub Actions, pytest, Git, Render, Vercel
- **Security:** prompt injection testing, OAuth 2.0, role-based access control, rate limiting, input validation, Metasploit, Aircrack-ng

## PROJECTS

### NBA Live Win Probability | Python, scikit-learn, XGBoost, FastAPI, Kafka

Jul 2026 - Sep 2026

- Built and deployed a live win-probability model that updates on every possession, trained on 6,149 games and validated on 4,500 games it had never seen.
- Selected the production model from eight candidates, cutting prediction error 9% and improving every accuracy measure tracked.
- Unified training and live serving on a single feature pipeline, so production predictions match tested behavior exactly.

### Governed RAG Assistant, UMass Boston Sustainable Solutions Lab | Python, ChromaDB, Gemini

Feb 2026 - May 2026

- Took a research lab's public AI assistant from 175 to 205 correct out of 208 on a graded benchmark scoring accuracy, source citations, and invented answers.
- Hand-checked all three remaining misses against the source documents and confirmed every one was a grading error rather than a wrong answer, leaving zero verified defects.
- Rebuilt search across 7,694 document sections by combining three retrieval methods, so questions phrased in unexpected ways still reach the right source.
- Held quality steady release to release with a 416-question regression benchmark run on every change.

### F1 Race Strategy Predictor | Python, scikit-learn, XGBoost, GitHub Actions

Dec 2025 - Mar 2026

- Built a predictor for the full race finishing order, evaluated across 8 seasons and roughly 157 races.
- Designed a leak-free evaluation that scores each race using only earlier ones, giving the team accuracy numbers they can trust and a clear read on which model actually performs best.
- Automated data refresh, feature selection and retraining every six hours, keeping predictions current with no manual work.

### Receipt Automation (AutoTax) | Python, LangGraph, FastAPI, Gemini

Jul 2025 - Sep 2026

- Built a pipeline that pulls receipts from email and files them automatically, sending only uncertain ones to a person: 100% correct routing across 16 receipt formats at 1.3 seconds each.
- Made processing 8x faster and cut AI usage roughly 70% by batching requests and trimming inputs, with no loss in accuracy.
- Backed every release with automated suites for accuracy, speed and security, including adversarial tests that try to trick the AI with malicious input.

### ClaimSight: Insurance Damage Assessment | FastAPI, Gemini vision, Docker

Jun 2026 - Sep 2026

- Built a tool that reads vehicle damage photos, prices the repair against live market listings, and produces an adjuster-ready summary.
- Added automatic quality scoring that grades every assessment and upgrades the weak ones before a person reviews it.

## EXPERIENCE

### VITA Program (IRS Volunteer Income Tax Assistance) | IRS-Certified Tax Preparer | Boston, MA

Sep 2021 - May 2023

- Earned IRS VITA certification and prepared 30+ federal returns for low-income filers, checking source documents before the program's required quality review.

### VietAID | Youth Leader | Boston, MA

Jul 2022 - Sep 2022

- Designed and ran mental health surveys for 200+ Boston Public School students, analyzed the responses, and presented recommendations to school officials.

### Gyu-Kaku Japanese BBQ | Server | Boston, MA | 20+ hrs/week alongside a full course load

Jul 2024 - Present